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Financial Mathematics Paper 1

QUESTION 1 (BASIC INTEREST + DEPRECIATION)

- 1.1. What is the effective annual interest rate for a nominal rate of 13,5% p.a. compounded quarterly?
- 1.2. What interest rate, compounded monthly, gives an effective annual interest rate of 17,3% p.a.?
- 1.3. A car is purchased today for R189 000. What will the book value of the car be after 5 years, if the depreciation rate is 16% p.a. calculated on a straight-line method. (3)
- 1.4. Nelly plans to have R25 000 at the end of 10 years. If the bank offers 8% p.a. interest compounded monthly, what amount should be deposited as a lump sum now so that she will have the required amount after 10 years. Assume that no further transactions take place on this account. (3)
- 1.5. Tembi buys a new car for R245 000. The value of the car depreciates at 13% per annum according to the reducing-balance method. After how many years will the value of the car be R83 543? (4)

QUESTION 2 (PRESENT VALUE + FUTURE VALUE ANNUITIES - BASIC)

- 2.1. You want to buy a new car. If you can afford to pay R5000 per month for the next 4 years and interest is calculated at 16,4% p.a. compounded monthly, determine the value of the loan you can afford. (4)
- 2.2. Refer to question 2.1. If you had rather saved the R5000 per month over a 4 year period, with a bank offering 6,8% p.a. compounded monthly, how much money would you have accumulated? (4)
- 2.3. You want to have R1 000 000 in the bank in 5 years' time. Interest is calculated at 5,3% p.a. compounded monthly. What monthly payment must you pay at the end of each month to achieve your goal?

2.4. Peter wants to buy a house costing R1250 000. He is required to pay 10% deposit and borrow the rest from the bank as a loan. The bank offers interest at 7,5% p.a. compounded monthly and expects the loan to be repaid over 20 years.

2.4.1. What amount does he have to borrow? (1)

2.4.2. What is Peter's monthly installment? (5)

2.4.3. What is the total amount that Peter will have paid on the loan after 20 years? (1)

2.4.4. How much in interest will Peter pay in total after 20 years? (2)

QUESTION 3 (OUTSTANDING BALANCE + LAST PAYMENT)

3.1. Mr Williams buys a house for R450 000. He pays a deposit of 10% and takes out a bank loan for the balance.

3.1.1. Calculate the value of the loan. (2)

3.1.2. He pays back the loan by means of equal monthly instalments over a period of 20 years. The first payment is made one month after the allocation of the loan. Interest is calculated at 8% per annum, compounded monthly.

Calculate the value of the monthly instalment. (4)

3.1.3. He decides to settle the loan after 17 years.

Calculate the outstanding balance on the loan if the last payment is made at the end of the 17th year. (3)

3.2. You have to pay off a loan of R400 000.
Interest is calculated at 18% p.a. compounded monthly.

If you decide to pay back R10 000 per month

3.2.1. how many payments will you have to make? (6)

3.2.2. what will the value of your last payment (< R10 000) be? (5)

QUESTION 4 (SINKING FUNDS)

In order to replace equipment, bought for R50 000, in 5 years' time, you invest money in a sinking fund, paying 12% p.a. compounded quarterly.

Deposits are made into the sinking fund on a quarterly basis.

The first payment is made immediately and the last payment after exactly 5 years.

Depreciation is estimated at 10% p.a. (reducing balance) and inflation at 7% p.a.

The equipment will be sold at its depreciated value after 5 years and new equipment will be bought from this money plus the money in the sinking fund.

4.1. What amount must be covered by the sinking fund in 5 years' time? (4)

4.2. What will the value of the quarterly deposit into the fund be? (5)

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QUESTION 5 (DEVIATIONS FROM NORMAL ANNUITY PATTERN)

5.1. John has to pay off a loan of R200 000, by means of monthly payments. He makes his first payment six months after the day the loan was made. He makes his last payment exactly 5 years after the day the loan was made. Interest is calculated at 13% p.a. compounded monthly. What is the value of his monthly payment?

5.2. You wish to pay off a loan of R100 000, by means of 24 quarterly payments, starting three months after the loan was taken. The interest rate is 12,7% per annum, compounded quarterly. The bank calculates that the value of your quarterly payment will be R6016,61.

5.2.1 Determine the outstanding balance on your loan after 9 payments. (4)

5.2.2 You miss the 10th, 11th and 12th payments. Three months after the 12th payment, you continue paying, but now the payment has to be adjusted in order to still pay off the loan in the original time. What is the value of the adjusted quarterly payment? (5)

Financial Mathematics: Solutions

Question 1:

$$1.1. \quad 1 + i_{\text{eff}} = \left(1 + \frac{i_{\text{nom}}}{n} \right)^n$$

$$1 + i_{\text{eff}} = \left(1 + \frac{0,135}{4} \right)^4$$

$$i_{\text{eff}} = 0,14199$$

$\therefore$ Rate 14,20%

$$1.2. \quad 1 + i_{\text{eff}} = \left(1 + \frac{i_{\text{nom}}}{n} \right)^n$$

$$1 + 0,173 = \left(1 + \frac{i_{\text{nom}}}{12} \right)^{12}$$

$$1 + \frac{i_{\text{nom}}}{12} = \sqrt[12]{1,173}$$

$$\frac{i_{\text{nom}}}{12} = \sqrt[12]{1,173} - 1$$

$$i_{\text{nom}} = 12 \left(\sqrt[12]{1,173} - 1 \right)$$

$$i_{\text{nom}} = 0,16063$$

$\therefore$ Rate 16,06%

$$1.3. \quad A = P(1 - i \cdot n)$$

$$= 189000(1 - 0,1615)$$

$$= \underline{\underline{R37800}}$$

$$1.4. \quad A = P (1+i)^n$$

$$25000 = P \left(1 + \frac{0,08}{12} \right)^{120}$$

$$P = \frac{25000}{\left(1 + \frac{0,08}{12} \right)^{120}}$$

$$P = \underline{R 11\,263,09}$$

$$1.5. \quad A = P(1-i)^n$$

$$83\,543 = 245\,000 (1 - 0,13)^n$$

$$\frac{83\,543}{245\,000} = 0,87^n$$

$$245\,000$$

$$n = \log_{0,87} \frac{83\,543}{245\,000}$$

$$n = 7,23$$

$\therefore \approx 7$ years and 9 months.

Question 2:

$$2.1. \quad P = \frac{x [1 - (1+i)^{-n}]}{i}$$

$$P = \frac{5000 [1 - (1 + \frac{0,164}{12})^{-48}]}{0,164/12}$$

$$P = \underline{R 175\,158,59}$$

$$2.2. \quad F = \frac{x [(1+i)^n - 1]}{i}$$

$$F = 5000 \left[\left(1 + \frac{0,068}{12} \right)^{48} - 1 \right]$$

$$F = \underline{R 274\,923,20}$$

2.3

$$F = \frac{x \left[(1+i)^n - 1 \right]}{i}$$

$$1000000 = x \left[\frac{\left(1 + \frac{0,053}{12}\right)^{60} - 1}{0,053/12} \right]$$

$$x = 1000000 \div \left[\frac{\left(1 + \frac{0,053}{12}\right)^{60} - 1}{0,053/12} \right]$$

$$\underline{x = R\ 14\ 592,32}$$

2.4.1.

$$1250\ 000 \times 0,9 = \underline{R\ 1125\ 000,}$$

2.4.2.

$$P = \frac{x \left[1 - (1+i)^{-n} \right]}{i}$$

$$1\ 125\ 000 = x \left[\frac{1 - \left(1 + \frac{0,075}{12}\right)^{-240}}{0,075/12} \right]$$

$$x = 1125\ 000 \div \left[\frac{1 - \left(1 + \frac{0,075}{12}\right)^{-240}}{0,075/12} \right]$$

$$\underline{x = R\ 9062,92,}$$

2.4.3.

$$\begin{aligned} &9062,92 \times 240 \\ &= \underline{R\ 2175\ 100,80,} \end{aligned}$$

2.44

$$\begin{aligned} &2175\ 100,80 - 1125000 \\ &= \underline{R\ 1050\ 100,80,} \end{aligned}$$

Question 3

$$3.1.1. \quad 450000 \times 0,9 \\ = \underline{R 405000},$$

$$3.1.2. \quad P = x \frac{[1 - (1+i)^{-n}]}{i}$$

$$405000 = x \frac{[1 - (1 + \frac{0,08}{12})^{-240}]}{\frac{0,08}{12}}$$

$$x = 405000 \div \frac{[1 - (1 + \frac{0,08}{12})^{-240}]}{\frac{0,08}{12}}$$

$$\underline{x = R 3387,58},$$

3.1.3 Outstanding balance =

$$405000 \left(1 + \frac{0,08}{12}\right)^{204} - \frac{3387,58 \left[\left(1 + \frac{0,08}{12}\right)^{204} - 1 \right]}{\frac{0,08}{12}}$$

$$= \underline{R 108\,104,85},$$

$$3.2. \quad P = x \frac{[1 - (1+i)^{-n}]}{i}$$

$$400000 = \frac{10000 \left[1 - \left(1 + \frac{0,18}{12}\right)^{-n} \right]}{\frac{0,18}{12}}$$

$$\frac{3}{5} = 1 - \left(1 + \frac{0,18}{12}\right)^{-n}$$

$$\frac{203}{200}^{-n} = \frac{2}{5}$$

$$-n = \log_{203/200} \frac{2}{5}$$

$$n = - \log_{203/200} \frac{2}{5}$$

$$n = 61,54$$

$\therefore$ 62 payments

$$3.2.2. \quad 400000 = 10000 \left[\frac{1 - \left(1 + \frac{0,18}{12}\right)^{-61}}{0,18/12} \right] + x \left(1 + \frac{0,18}{12}\right)^{-61} \quad (5)$$

$$400000 = 397835,16 + x \left(1 + \frac{0,18}{12}\right)^{-62}$$

$$2164,84 = x \left(1 + \frac{0,18}{12}\right)^{-62}$$

$$x = 2164,84 \left(1 + \frac{0,18}{12}\right)^{62}$$

$$\underline{x = R\ 5449,04,}$$

Question 4

$$4.1. \quad A = P(1+i)^n \quad (\text{new equipment})$$

$$A = 50000(1+0,07)^5$$

$$= \underline{R\ 70\ 127,59,}$$

$$A = P(1-i)^n \quad (\text{old equipment})$$

$$= 50000(1-0,1)^5$$

$$= \underline{R\ 29\ 524,50,}$$

$$\text{Sinking Fund} = 70127,59 - 29524,50$$

$$= \underline{R\ 40\ 603,09,}$$

$$4.2. \quad F = \frac{x \left[(1+i)^n - 1 \right]}{i}$$

$$40603,09 = \frac{x \left[\left(1 + \frac{0,12}{4}\right)^{21} - 1 \right]}{0,12/4}$$

$$x = 40603,09 \div \frac{\left[\left(1 + \frac{0,12}{4}\right)^{21} - 1 \right]}{0,12/4}$$

$$\underline{x = R\ 1415,90,}$$

Question 5

$$5.1. \quad 200000 \left(1 + \frac{0,13}{12} \right)^5 = x \left[\frac{1 - \left(1 + \frac{0,13}{12} \right)^{-55}}{\frac{0,13}{12}} \right]$$

$$x = 200000 \left(1 + \frac{0,13}{12} \right)^5 \div \left[\frac{1 - \left(1 + \frac{0,13}{12} \right)^{-55}}{\frac{0,13}{12}} \right]$$

$$= \underline{R 5113,97},$$

5.2.1. Balance =

$$100000 \left(1 + \frac{0,127}{4} \right)^9 - 6016,61 \left[\frac{\left(1 + \frac{0,127}{4} \right)^9 - 1}{\frac{0,127}{4}} \right]$$

$$= \underline{R 70925,10},$$

$$5.2.2. \quad P = x \left[1 - (1+i)^{-n} \right]$$

$$70925,10 \left(1 + \frac{0,127}{4} \right)^3 = x \left[\frac{1 - \left(1 + \frac{0,127}{4} \right)^{-12}}{\frac{0,127}{4}} \right]$$

$$\therefore x = 70925,10 \left(1 + \frac{0,127}{4} \right)^3 \div \left[\frac{1 - \left(1 + \frac{0,127}{4} \right)^{-12}}{\frac{0,127}{4}} \right]$$

$$x = \underline{R 7907,72},$$